

Improving DNA quality extracted from fecal samples—a method to improve DNA yield

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Abstract The study and conservation of endangered species is an important topic but collecting genetic samples is challenging as many of the threatened species are rare, elusive, or difficult to approach. Thus, the use of noninvasive samples to obtain genetic information has been gaining popularity. The noninvasive samples generally yield low DNA quantity and quality, which interferes with the genetic analyses performed. Several studies have identified factors that influence the amount and integrity of DNA extracted from noninvasive samples. Our work introduces a noninvasive DNA extraction method that can be modified and adapted to recover a large amount of DNA from stool samples. The protocol described here consists of two digestions and two purification steps and was tested on samples collected in the field from wild animals. This new method proved to be effective in the DNA extraction from scat samples in different conditions and from different species. The concentration of extracted DNA ranged from 30 to 70 ng/μl for *Equus hemionus*, *Equus africanus*, and *Equus kiang*. The success of mtDNA amplification ranged between 80 and 100% while for microsatellite markers the rate was between 65 and 100%. The reproducibility between scorings of the amplified fragments ranged between 76 and 100%. The current method has successfully improved the DNA yield on several species of wild equids. In addition to the classic

genetic markers, we also test this method suitability for next-generation sequencing methods, by obtaining a RADseq library of 16 nM using 16 cycles and fragment size distribution similar to what is expected for high-quality samples. This protocol has successfully improved the DNA yield on different species by changing some steps of a classical spin-column DNA extraction protocol. We conclude that our fecal DNA extraction method can be used to provide DNA for a variety of downstream analyses.

Keywords Fecal samples · Equids · DNA extraction · Noninvasive sampling · Protocol · Next-generation sequencing

Introduction

Noninvasive methodologies have been widely applied to population genetic studies due to concerns regarding the disturbance of wild animals when both destructive and invasive sampling methods are used (Taberlet et al. 1999). Indeed, it has been suggested that all conservation genetic studies should utilize noninvasive samples (DeSalle and Amato 2004), since it is the only way to obtain genetic data from cryptic or endangered wildlife populations without disturbing the animals (Waits and Paetkau 2005). Because animals are not captured or handled, the risk of injuries and disturbing the behavior of the group are eliminated (Taberlet and Luikart 1999). Although almost any trace left by the animals can be used for DNA extraction (Beja-Pereira et al. 2009; Waits and Paetkau 2005), scat is the most common source since it allows researchers to obtain not only genetic data but also other information such as physiological hormones, diet, and gut parasite load (Luikart et al. 2008).

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The host DNA present in fecal samples, which is the most common source for noninvasive sampling, is mostly derived from the intestine epithelial cells. Therefore, the extracted DNA is usually degraded, in very low amount and with a high prevalence of inhibitors (Taberlet et al. 1999) which causes the limitations already described that are associated with this methodology.

In the field, fecal samples are collected in an opportunistic way and sometimes the animals are not even observed, which can make the estimation of the time the samples were left exposed on the field challenging. The exposure time and the environmental conditions, e.g., temperature and moisture, govern the success of DNA extraction, and in the field, those conditions are not under the control of the researchers. Even though several conditions were under control, previous reports reveal that the success of the study depends on every aspect of it, starting with the time and type of sample collection, the season of sampling, diet, methods of extraction, and interaction between all of those factors (Beja-Pereira et al. 2009; Frantzen et al. 1998; Maudet et al. 2004; Piggott and Taylor 2003; Soto-Calderón et al. 2009). Comparison studies indicate that diet and environmental conditions are the two major factors affecting the quality and quantity of the extracted DNA.

The main goal of the current work was to develop a method that provides high amounts of DNA and can be used in several species, with different ecological conditions. In our case, we have worked with several wild equid species with flexible and adaptable diets according to the available resources, which range from grasses to trees (Duncan et al. 1992; Schulz and Kaiser 2012). The differences in the digestive tract of equids, when compared to other grazing species, may result in different success rates when similar methods are adapted to different species. As most of the wild equids are threatened and difficult to approach because of their behavior and habitat (e.g., desert, savannah, high altitude), the development of a suitable noninvasive DNA extraction protocol is crucial to collect genetic information from these species.

Noninvasive samples are sensitive and often do not provide reliable results when standard methodologies are applied. Essentially, we cannot adapt our samples to our existing protocol but we reasoned that we may achieve better results if we adapt the protocol to the samples. The DNA extraction method presented in the current work is different from the previously published protocols due to (i) the use of two digestion steps, (ii) the fact that the second digestion occurs after a purification of the initial extract, and (iii) during the entire process; several steps may be modified based on individual sample characteristics. The uniqueness of this method relies on its dynamics and only requires basic knowledge of the sample that is being processed (e.g., diet and environmental conditions) in order to adapt the steps to the type of sample. It is widely accepted that noninvasive sampling has three main

problems: DNA degradation, DNA quantity, and DNA quality. It is not possible to overcome pre-existing DNA degradation; however, DNA quantity and quality issues can be solved. This method focuses on improving what can actually be enhanced, by maximizing DNA yield and minimizing the inhibitory effects.

Usually, when the animals' diet is rich in easily digestible fibers, the digestion process will be faster and less abrasive to the intestine epithelium, culminating on a lower amount of epithelial cells. Therefore, it is important to have some previous knowledge of the subject species/population diet. This can actually be achieved by simple observation of the available food resources on the area where the samples are being collected. Another ecologic characteristic that should be taken into account is whether the sample was exposed to UV (e.g., when the collection site is in high altitude and deserts) as this will also influence the quality of the extracted DNA. This knowledge is required at the first step of our method in order to determine the amount of sample to use. That amount of fecal material is increased when we know (or assume) that the animals' diet is rich in digestible fiber and/or the sample is possibly degraded due to harsh environmental condition exposure.

There are four main steps that can be changed to increase the total amount of extracted DNA. The first one is during the first digestion when the total amount of used sample should be increased in certain situations, as described above. The second is immediately before the first purification step when evaporation is used to concentrate the DNA. This evaporation step is not crucial when the samples are well preserved; however, it is of extreme importance in poorly preserved samples (e.g., samples exposed to moisture and not dried immediately after collection), degraded samples (e.g., exposed to UV rays or collected a long period after defecation), and when the diet is rich in fibers. The third step can then be added between the first purification and the second digestion. In this step, after the initial purification, the samples are transferred to a new tube where the second digestion will occur. With well-preserved samples, only one tube is necessary; but with poor-quality samples, multiple tubes (replicates) are advisable. When using replicates, the amount of DNA retrieved will be maximized, even with high fiber content diet or exposure to harsh environmental conditions. Also, in this same step, another change can be made instead of using the recommended standard lysis buffer; CTAB buffer should be used if the diet is rich in fiber and/or if the sample is poorly preserved (e.g., if it was exposed to moisture and presents, for instance, fungi on the superficial layer). The digestion step itself should also be carried at a lower temperature and for a longer period of time so that DNA amount will be maximized. The last main step where alterations should be carried is the second purification. Usually, noninvasive sample extracts are not transparent and the color may vary between light and dark. Darker extracts

usually carry a higher amount of inhibitors; therefore, the purification steps should be repeated in order to obtain a clear color before the final elution step.

The present method was initially designed for the African wild ass since neither previously published methods nor commercially available kits allowed the recovery of sufficient DNA to perform genetic analyses. From the original protocol, several steps can be altered in order to apply this same protocol to other species or samples in different conditions. The methodology is based on a double digestion, for which the second digestion is performed after a purification step which allows the removal of a large part of the inhibitors. Our main goal was to provide a new extraction method that could be applied to different species and to samples with variable environmental conditions, collected on the field and without any sort of control. In addition, we also intended that the method could be applied when it is not possible to extract DNA immediately after collection, which is actually a situation that often occurs in the field, as for instance, sampling in deserts or in high-altitude plateaus, which are commonly far from a laboratory. This method can successfully purify DNA from samples collected in field conditions.

Material and methods

Sample collection

Equids are characterized as fermenting herbivores that are well adapted to low-quality and fibrous food and are spread through steppes, semi-arid, and arid environments. Although extant equids are closely related species, each of them has a particular range of diet sources as a result of different climatic conditions (Schulz and Kaiser 2012). The use of three equid species, with different environmental characteristics, allows us to verify the applicability of the current method to samples under different conditions. As these species are closely related, it is possible to compare the obtained results and infer the influence of climate, diet, and sample collection, on the success of DNA extraction. Another important factor that is often a source of problems is the large size and the nonpellet form of the feces, on which is necessary to select a region of the outer surface for the extraction process. Normally, small pellet-like feces are easy to surface wash in a 15-ml plastic tube with lysis buffer.

The samples had different conservation status and 20 samples, per species, were used. *Equus africanus* samples were stored for 2–3 years at room temperature but were dried immediately after collection; *Equus hemionus* samples were also dried immediately after collection and stored for 6 months at room temperature; *Equus kiang* samples were collected in the winter and were dried upon arrival at the laboratory, where they were dried and stored for 6 months at room temperature

before the DNA extraction. All of the samples were stored at room temperature, in bags containing silica gel, after ensuring that they were totally dry (if not, the samples were placed in an incubator at 40 °C until the moisture was removed and the samples were totally dry).

DNA extraction

To prevent contamination during the extraction process, all procedures were carried out in a laminar flow chamber and all the equipment used were sterilized with bleach and exposed to UV light before and after usage. The extraction bench and the PCR room are physically separated in order to avoid contamination between steps. The extraction method may be altered at several points, depending on the sample characteristics, in order to maximize the DNA amount obtained (Fig. 1). In each set of samples, a negative control tube, containing everything except the sample, was included in order to detect potential contamination.

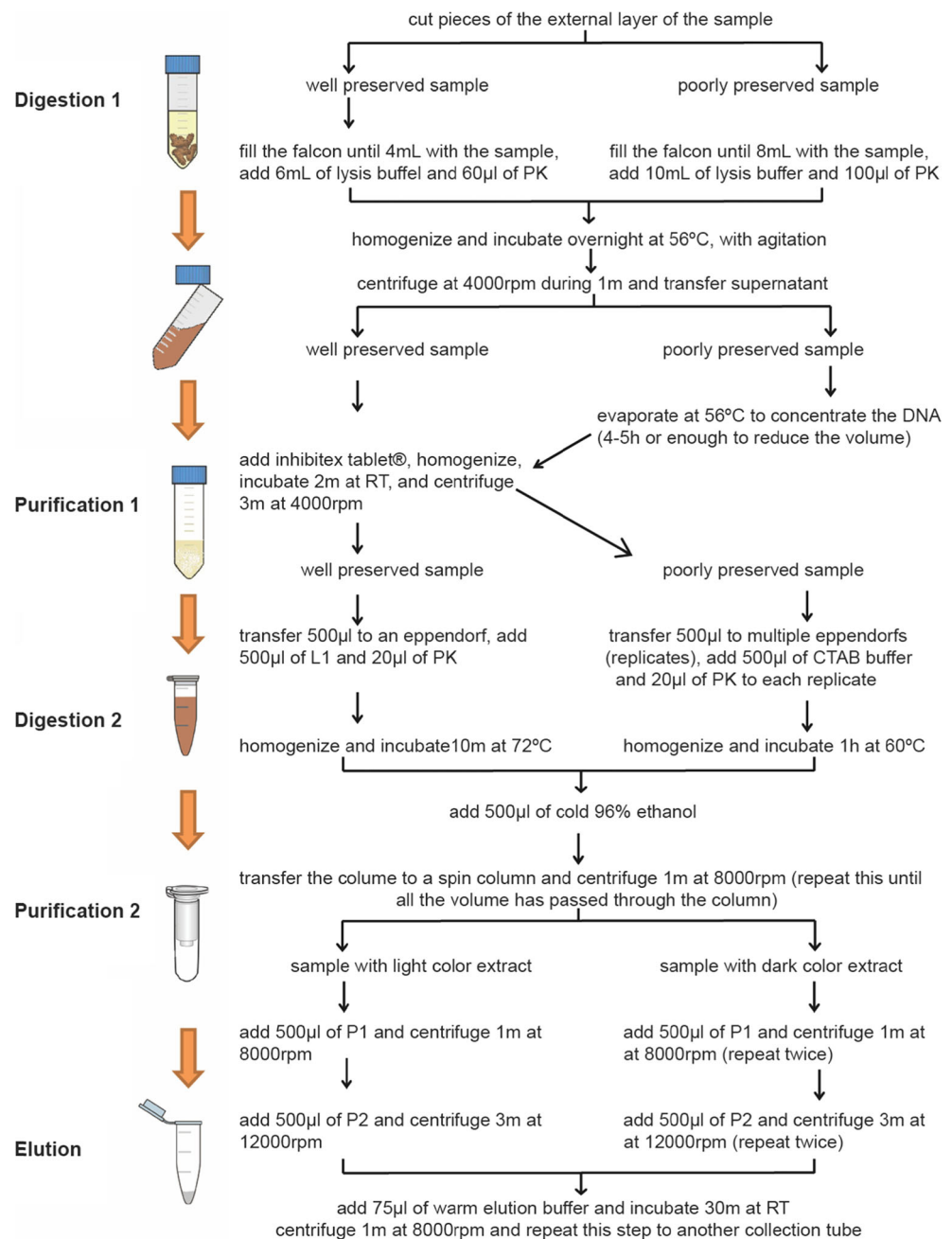
First digestion

Superficial portions of the stool samples were removed using a sterile scalpel and placed in 15-ml falcon tubes; this step used only a shallow slice of the fecal sample in order to prevent the inclusion of inhibitors contained in the interior part of the sample, and the scalpel was sterilized with bleach and washed twice with 96% ethanol and flamed between samples. Ideally, the portion of the sample that has been least exposed to UV rays, i.e., the stool that was facing the ground and that generally is darker should be selected. The amount of sample transferred to the tube depends on the sample size and condition; for samples poorly preserved (e.g., containing fungus and from animals with high fiber content diet), the amount should be increased. The falcon tubes were filled to 10 ml (or enough to cover the samples) with lyses buffer composed of 0.5 M Tritonplex, 0.1 M EDTA, and 2% sodium dodecyl sulfate. Between 50 and 100 µl of Proteinase K (20 mg/ml) was added to the tubes, depending on the amount of lyses buffer in the tube, keeping a proportion of 1:100 of Proteinase K and lyses buffer. The samples were immediately agitated by vortexing and incubated at 56 °C overnight, with mild agitation.

First purification

The tubes were centrifuged at 4000 rpm for 3 min. The supernatant was transferred to a new 15-ml tube leaving any solids behind. An additional step may be added here; in order to concentrate the DNA, these new tubes can be placed in an incubator, at 56 °C, opened, and covered with paper until part of the volume evaporates, so the DNA will be concentrated. This extra step is especially important if the samples may have

Fig. 1 Schematic representation of the steps involved in the noninvasive sample extraction. The steps were followed according to the different conditions of the samples (e.g., poor preservation or presence of inhibitors). *L1* commercial blood lysis buffer, *P1* commercial first purification reagent, *P2* commercial second purification reagent



a low amount of cells (e.g., if the animal diet is very rich in fiber, the number of epithelial cells will be reduced due to low abrasion on the intestine), otherwise this step may be skipped. After partial evaporation or centrifugation, depending on the sample, an inhibitEX® tablet (QIAGEN, GmbH, Hilden) was added to each tube and homogenized via vortex until the tablet was fully dissolved. The tubes were incubated at room temperature for 2 min and then centrifuged at 4000 rpm for 3 min. Next, 500 µl of the solution was transferred to a 1.5-ml Eppendorf tube. In this step, several replicates can be made by using multiple Eppendorf tubes. Once again, in samples with poor conditions, it is advisable to perform replicates to

maximize the DNA yield. After this step, any commercial kit can be used as long as it is suitable to extract blood samples.

Second digestion

To each tube, 20 µl of Proteinase K (20 mg/ml) and 500 µl of blood lysis buffer were added. If the animal diet consists of very high fiber content or if it has very strong inhibitors (e.g., fungus), the blood lysis buffer should be replaced by 500 µl of previously warmed (60 °C) CTAB buffer (2% CTAB, 1.3 M NaCl, 0.2% 2-Mercaptoethanol, 20 mM EDTA, 100 mM Tris-HCl). Either way, the tubes were homogenized in the vortex

for 15 s and incubated at 70 °C for 10 min (for very well-preserved samples) or 60 °C for 1 h (poorly preserved samples).

Second purification

Tubes were removed from the incubator and 500 µl of cold 96% ethanol was added and the solution was homogenized. Six hundred fifty microliters of the solution was then transferred to a DNA purification column and centrifuged at 8000 rpm for 1 min; this procedure was repeated until all the volume of each tube passed through the column. After the extract passed through the column, 500 µl of the commercial first purification buffer was added and the tubes were centrifuged at 8000 rpm for 1 min, and this step was repeated one more time if the samples had a very dark color. Finally, 500 µl of commercial second purification buffer was added and the columns were centrifuged at 12000 rpm for 3 min (this step should be repeated if the extract is a dark color); after this step, if the column still retained some solution, the tubes were centrifuged 2 additional minutes at 12000 rpm.

Elution

To each column, 75 µl of pre-heated (70 °C) commercial elution buffer was added and the tubes were incubated at room temperature for, at least, 30 min and centrifuged at 8000 rpm for 1 min. This elution procedure was repeated one additional time using a new collection tube, creating two isolates for each sample.

DNA quantification, amplification, and genotyping

Quantification

All the samples were run on a 0.8% agarose gel and quantitated with a Qubit® fluorometer (Invitrogen™), according to manufacturers' protocol. Samples with unusually high DNA concentrations (above 100 ng/µl) were also quantified using NanoDrop™ (Thermo Scientific), following the manufacturer's standard protocol, in order to confirm the amount of DNA present on the sample.

Equid DNA fragment amplification

All samples were amplified for one mtDNA (mitochondrial DNA) fragment of 450 bp (base pairs) and three microsatellites (COR20, COR90, NVEQH18) with sizes ranging from 100 to 200 bp, as described previously (Rosenbom et al. 2015).

Sample genotyping

After PCR, all samples were run on a 2% agarose gel to verify amplification. Prior to sequencing and loci genotyping, the amplified DNA fragments were diluted to obtain similar concentrations. mtDNA sequences were corrected with DNASTAR 6.0 software (DNASTAR Inc., Madison WI, USA) and compared with previously published fragments to ensure that sequences corresponded to target DNA. For microsatellites, GeneMapper v4.0 (Applied Biosystems, USA) software was used and all samples were independently scored by two users, to ensure reliable readings.

For the mtDNA amplification, only one replicate of each sample was used, while in microsatellites, two replicates of each sample were amplified. The scoring coherence was assayed by determining if both replicates yielded the same genotype or not, in one single amplification and scoring trial.

RADseq library preparation

In order to determine the potential application of this DNA extraction method to next-generation sequencing techniques, a RAD (restriction site-associated DNA) library was prepared. We followed a new RAD protocol using biotinylated RAD adaptors (Ali et al. 2016) using the SbfI enzyme, using 20 ng of DNA per individual.

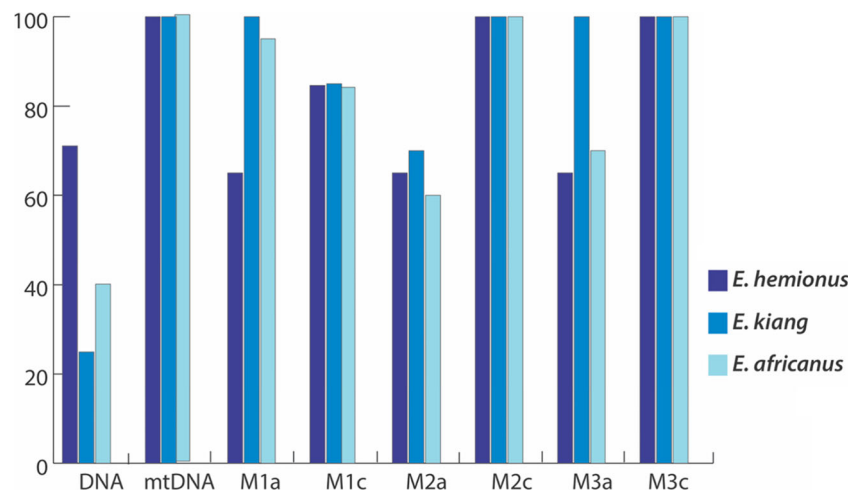
Results

DNA was successfully recovered from all of the samples. The *E. hemionus* was the species yielded a higher DNA amount, with an average of 71.12 ng/µl, ranging from 7.39 to 179 ng/µl. The *E. africanus* samples produced DNA concentrations between 9.90 and 257 ng/µl, with an average of 40.16 ng/µl. *E. kiang* produced an average of 24.97 ng/µl, with individual values between 2.51 and 77.90 ng/µl.

Amplification of mtDNA was successful in every sample for all three species. Microsatellite amplification success was variable among markers and populations and ranged from 65 to 100% (Fig. 2). The coherency between scorings, i.e., both replicates yielded the same genotype, ranged between 76 and 100%. Additionally, no contamination was detected both in extraction and PCR negative controls.

For the RADseq libraries, although the samples were not sequenced, the results support that this extraction method may be used for future next-generation sequencing studies. We obtained a sequencing library of 16 nM using 12 cycles, and the distribution of the fragment sizes corresponded to what is obtained with invasive/high DNA quality samples (~150–500 bp).

Fig. 2 Comparison of the results obtained for the different species regarding DNA amount (DNA, ng/ μ l), percentage of success of mtDNA amplification (mtDNA), and percentage of amplification success (a) and scoring coherency in amplified samples (c), in all the three markers (M1 COR20, M2 COR90, and M3 NVEQH18)



Discussion

All of the equid scat samples analyzed in the present study were successfully extracted and yielded amplifiable DNA, either for mtDNA or microsatellites. However, it is noteworthy that variation was observed among samples and, also, between species. It has been reported that the same extraction procedure when used on different species may result in reduced amplifiable DNA (Taberlet and Luikart 1999). Numerous studies have shown the effect preservation methods have on noninvasive samples, suggesting that inadequate storage can result in reduced DNA quantity and quality (Frantzen et al. 1998; Murphy et al. 2002; Piggott and Taylor 2003; Roeder et al. 2004; Soto-Calderón et al. 2009; Waits and Paetkau 2005). The samples used in this study represent different species and were collected under very different environmental conditions. The population yielding a high concentration of DNA corresponded to *E. hemionus* which, even though was stored for 6 months prior to extraction, was dried immediately after collection. This population yielded an average of 71 ng/ μ l of DNA per sample. This result could be due to several factors. The impact of the diet on the amount of amplifiable DNA is known (Broquet et al. 2007; Piggott and Taylor 2003) since the target DNA present in fecal samples (with the exception of the diet contents itself) is from the intestine epithelial cells. Therefore, if the animals have a diet that is somehow more abrasive to the intestine, then we should be able to retrieve more DNA than when the animal has a different diet, e.g., richer in high digestibility fiber which decreases the digestion period. Another important factor is related to the season when the samples were collected, as it was already demonstrated the influence of the different development state of the food eaten by the herbivores, in terms of DNA quality extracted from noninvasive samples (Maudet et al. 2004). Indeed, *E. hemionus* samples were collected during the dry season, which not only permits the samples to dry

quickly, preserving the DNA, but also suggests that the available grass is drier and thus has a higher content of fibers with lower digestibility (as a result of the increase in lignin and cellulose as the plant matures), being therefore more abrasive and retaining more epithelial cells.

The *E. africanus* samples were also stored for a relatively long period (2 to 3 years) prior to extraction and this population yielded relatively high DNA amounts. Since these samples were collected in a very warm and dry environment, the high temperatures caused rapid fecal desiccation. This, combined with the animals' diet (probably with low digestibility fiber content, given the dry climate), produced samples preserved and stored in excellent conditions. Nonetheless, it must be noted that the high environmental temperatures associated with these samples also have a negative impact on the DNA quality as these temperatures are a consequence of high level solar ultraviolet light.

Although a reduced amount of DNA was achieved for *E. kiang*, when compared with the other two species, the quantities of DNA extracted at ~25 ng/ μ l were still sufficient for downstream analyses. The lower quantities obtained from these samples may be associated with the poor storage conditions on which these samples were kept for 6 months, until their arrival at the laboratory facilities. Another fact adding to this is that samples were collected during the winter, which is associated with higher humidity, and at a very high altitude, consequently with high UV exposure.

The mtDNA amplification was successful with all of the samples. To our best knowledge, there is no previous noninvasive study with equids that allow us to compare amplification rates; however, for other herbivores, such as the Asian elephant or reindeer, the mtDNA amplification rate is also close to 100% (Fernando et al. 2003; Flagstad et al. 1999), which was the success rate we found for our equid samples.

As expected, since mitochondrial DNA is present in higher amounts than nuclear, the success rates of nuclear DNA

amplification were, in average, lower. For some species, some markers had an amplification rate of 100%, while with other markers the same rate decreased, supporting the previous evidence that loci choice has a major role in the success of noninvasive studies (Broquet et al. 2007; Buchan et al. 2005). The average amplification was 65% for *E. hemionus*, 75% for *E. africanus*, and 90% for *E. kiang*. Regardless of the lower amplification success rate, *E. hemionus* samples had a scoring coherency of the amplified fragments of 95%, similar to *E. kiang*. This was measured only by one single amplification trial of two replicates and the presence or absence of similar genotypes, which allow us to hypothesize that with more amplification attempts, as recommended for noninvasive samples, this value would also increase.

Besides aspects such as diet and locus choice, there are other aspects that also determine success rates, such as field conditions. A recent study, reported that the success rate of amplification decreased to almost half when the fecal samples were in the field for longer periods than 24 h (85% success for samples exposed for less than 24 h, versus 44% for longer periods) (Barbosa et al. 2013). Obtaining field scat samples quickly after defecation is not always possible and sometimes it is difficult to estimate for how long samples were out on the field, exposed to adverse environmental conditions. That exposure will have consequences on DNA degradation that may compromise the study (Beja-Pereira et al. 2009). However, it may not be possible to change the scat collection itself since, for many threatened and cryptic species, samples are collected opportunistically. Therefore, the adaptation of the DNA extraction protocols is crucial for the success of all downstream genetic analyses. Here, we have demonstrated that a few changes in the protocols can increase the amount of DNA extracted from noninvasive samples of different species. Also, the RAD library results allow us to predict the viability of the current method for next-generation sequencing methodology. The protocol described here can be applied to a variety of field conditions and allows researchers to improve the quantity and quality of DNA. Besides being adaptable to field conditions, the present method is also affordable since it only requires standard equipment, reagents, and extraction kits. It is important to state that not all DNA recovered belong to the target species, as noninvasive stool samples will also yield nontarget DNA, as for instance bacterial DNA.

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Authors' contributions VC designed the protocol; VC, SMO, SR, and RM performed the laboratory experiments; VC, SMO, and AB-P wrote the paper and supervised the laboratory experiments.

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